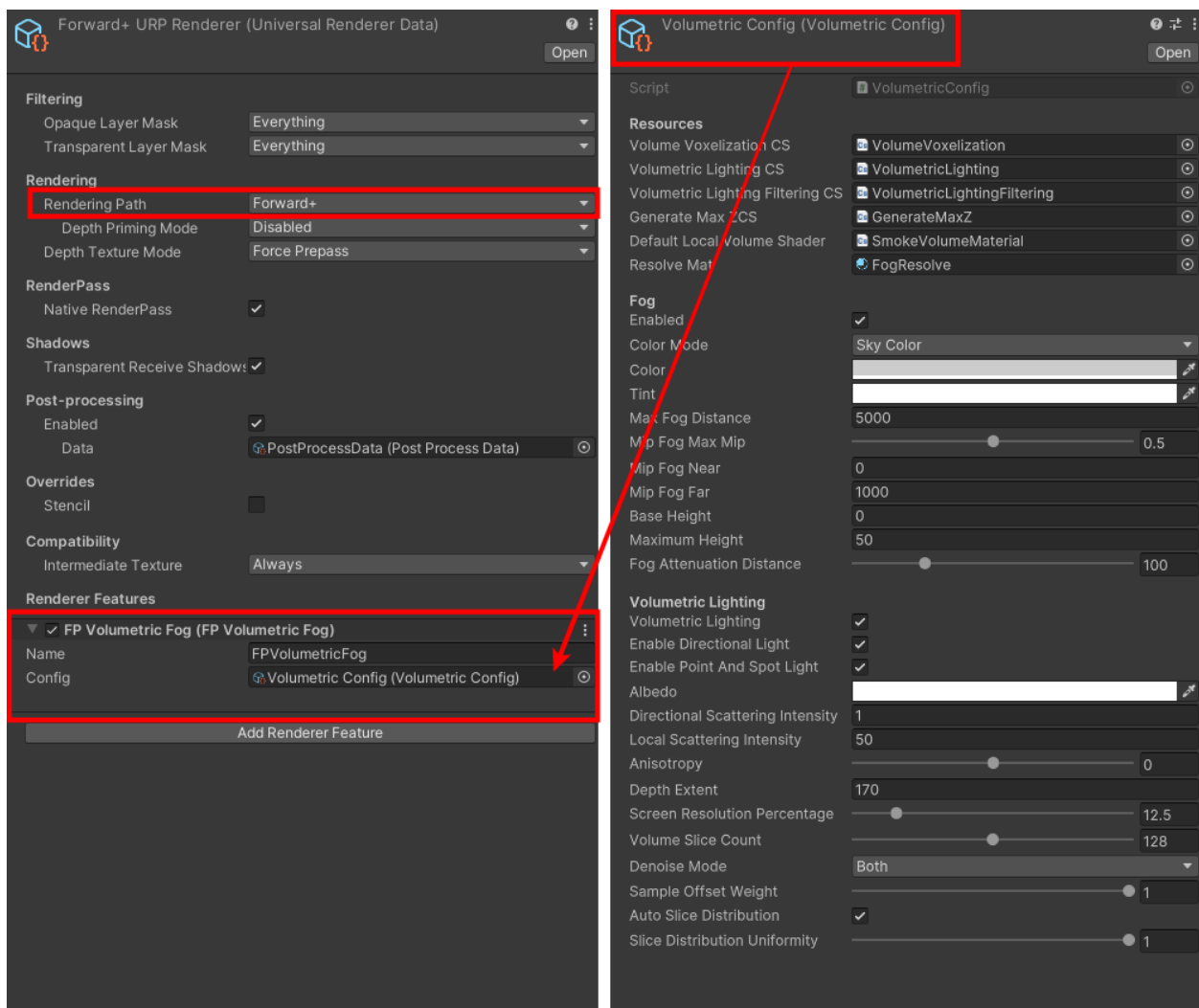


Ethereal 2024 v2.6

The new **Ethereal 2024** module adds a much faster method of handling local shadowed lights, especially fast in Forward+ and supporting all of Forward, Forward+ and Deferred modes. The **original Ethereal system in "VolumeFogSRP" folder should still be used for volumetric fog and best sun volumetric light control, plus special FX like haze and impostor volume lights**. This new system will be expanded with more functionality in next updates and the two systems can be used in combination.

How to Use

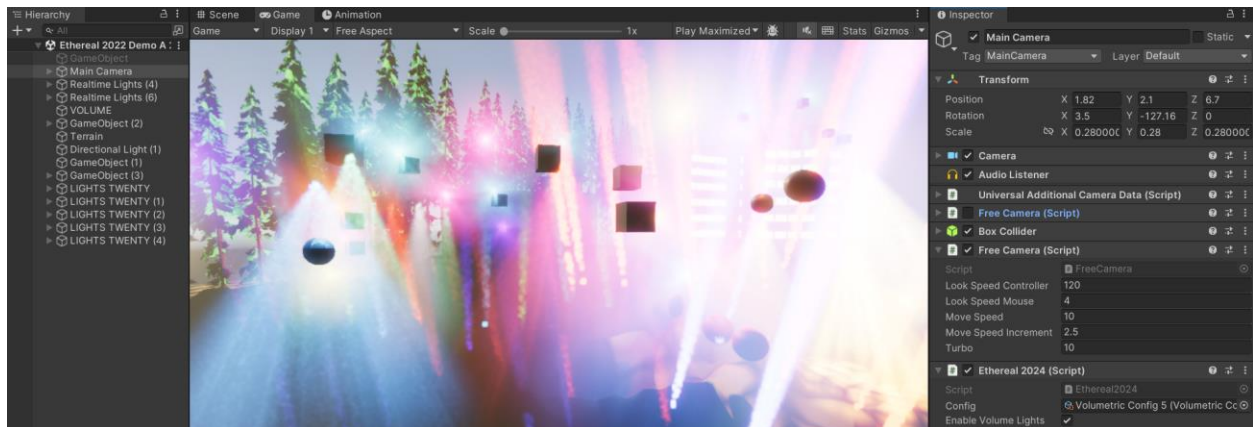
1. Add '**FP Volumetric Fog**' renderer feature to Renderer data
2. Create 'Volumetric Config' via 'Create/UniversalVolumetric/VolumetricFogConfig'
3. Link the 'Volumetric Config' asset to the renderer feature
4. **Forward+ rendering path is strongly recommended** for best performance gain



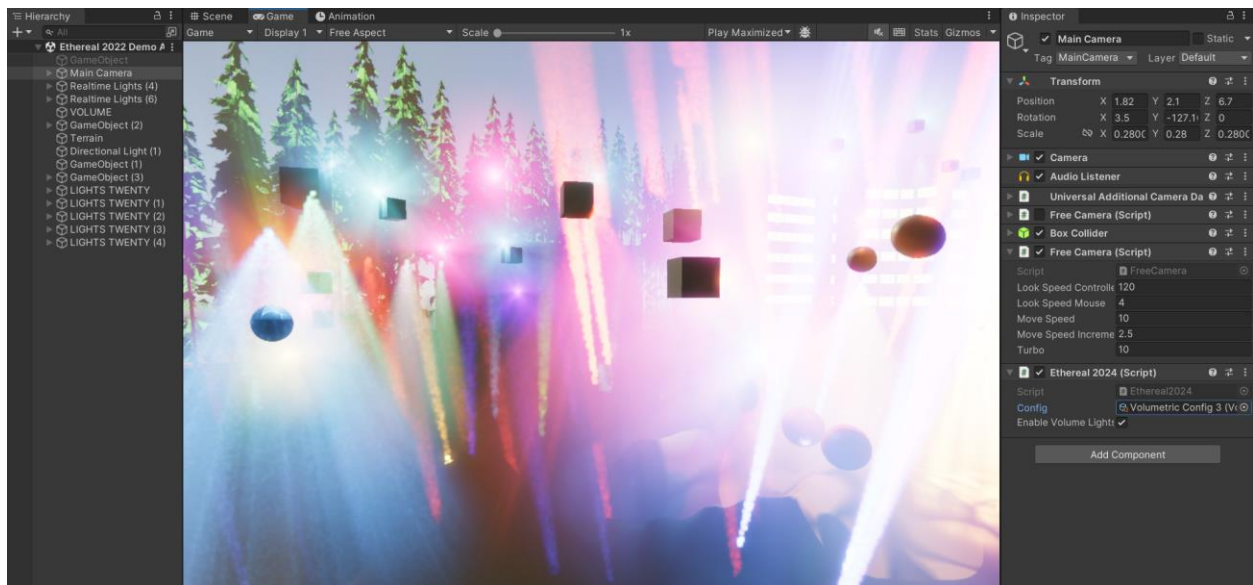
Ethereal 2024 controller

The system uses a controller script "**Ethereal 2024**" on the MainCamera that applies the user defined settings file and enables or disables the effect, such that can use a different setting per scene. The script must be placed on the **camera tagged as "MainCamera"** in the scene. It is recommended to use this script on every scene that requires the effect, so can use different settings per scene.

To disable the fog, uncheck the "Enable Volumetric Lights" checkbox or disable the script itself when it is still on the camera.



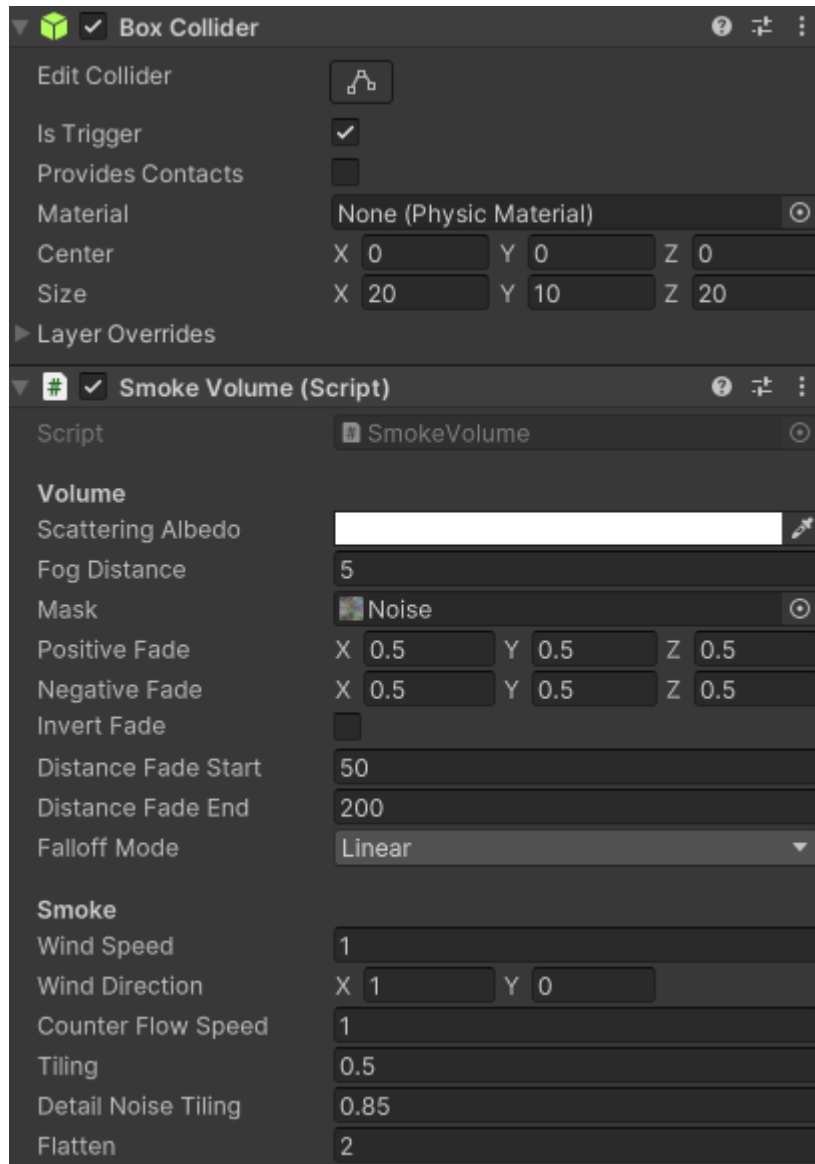
Example using another setting



Smoke Volume

The system assumes a global volume, and local smoke volumes can be used to make a ground smoke effect for specified areas. Smoke volume component uses a box collider to determine its area. Due to performance, 4 smoke volumes are supported currently.

To add a smoke volume to the scene, create a new gameObject and add the '**Smoke Volume**' component to the gameObject. A default noise texture is available, which can be found at 'Runtime/Textures/' directory.

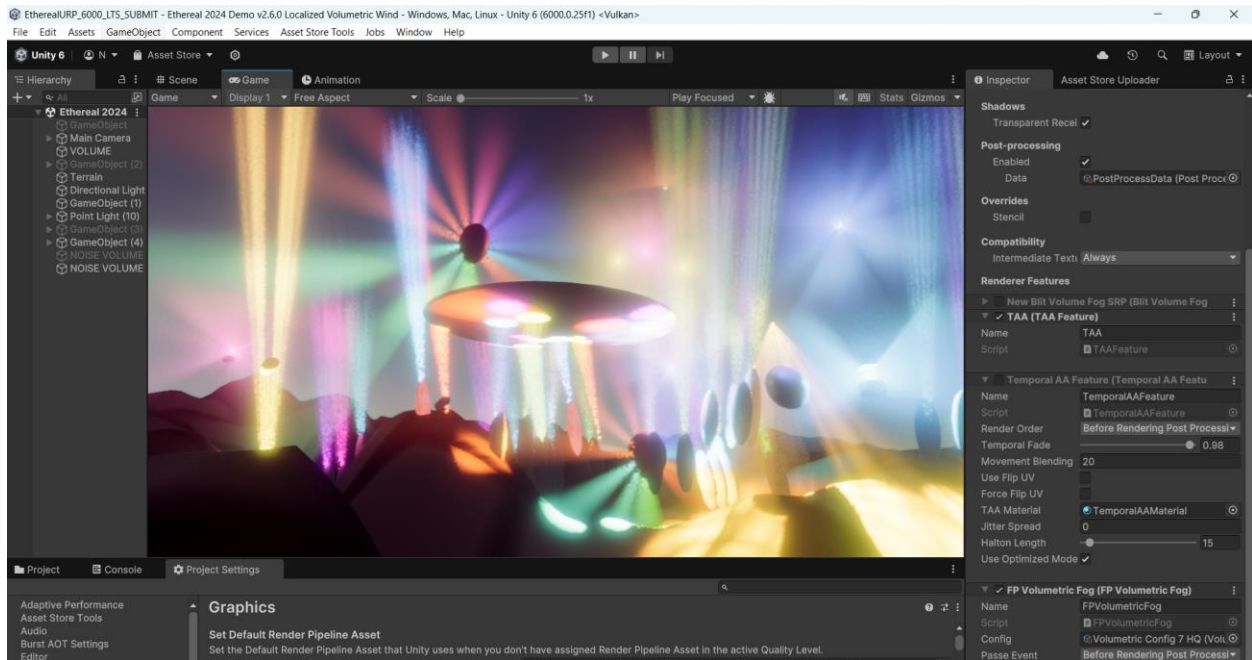


Compatibility

The system in Unity 6 RenderGraph is **compatible with DX11 for all modes**, for both Ethereal 2021 and Ethereal 2024 modules.

In Vulkan all modes are also supported, but special consideration has to be taken for the render order of the Ethereal 2024 effect, as shown below, in Before rendering Post Processing.

In DX12 the Ethereal 2021 is not compatible and only the Ethereal 2024 module described in this manual is available.



Known Issues

1. [Ethereal 2024](#) memory leak issue note in Unity 6:

If there is a memory leak in Unity 6 Preview when using the scene view in editor, while using the included sample ETHEREAL pipeline, please remove the first default renderer and replace it with one that does not have any image effects, as the scene view seems to have a bug that can increase RAM due to the default renderer have image effects. Can use any other renderer for the game camera, than the default first renderer.

This issue only affects Unity 6 and only when in Editor, in build there is no issues. Also it is generally recommended to avoid using image effects in the first default renderer, as helper cameras in general may try to use those effects.

Note that the Default renderer without image effects can be in any place in the pipeline and not necessarily in the first slot, by setting the relevant renderer in the renderers list as default.

2. Screen going black at certain camera rotations

If the Camera has both rotation and X,Y,Z coordinates at zero, the screen may go black, to fix add a tiny displacement in Z axis of camera (0.0001 will do). The issue has been fixed in v1.1.8c.

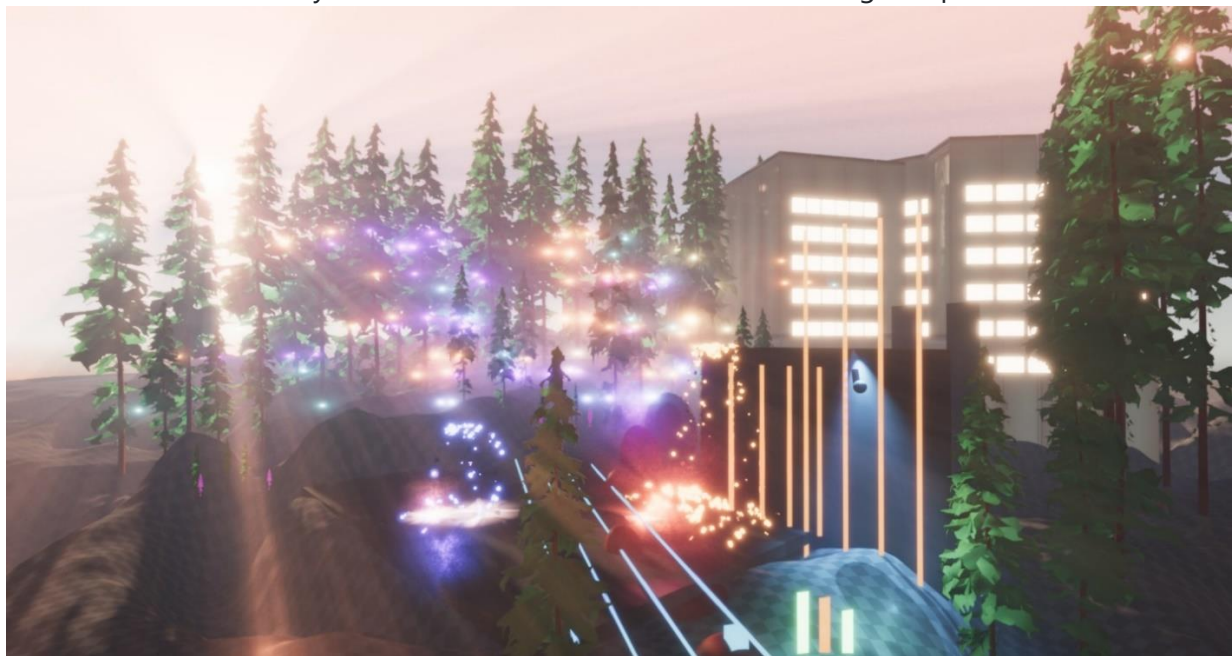
Performance

The system has a **much increased performance in shadowed point and spot volumetric lights** versus the original Ethereal module, especially if the Forward+ mode is used. The **Forward+ is the suggested mode** and is now fully supported by the other advanced lighting ARTnGAME asset [LUMINA real time global illumination](#), so both systems can be combined under that rendering mode.



Use with Ethereal Fog and impostors system

The main Ethereal system allows fog and impostors lights, that might want to enable extra to the new Ethereal 2024 system that is focused on local shadowed lights optimization.



Use with Sky Master ULTIMATE Volumetric Clouds

The system is also fully compatible with and included in the latest Beta 26 of [Sky Master ULTIMATE](#) asset, for combination with advanced weather and volumetric clouds.

Use with GLAMOR Image FX stack

The system is fully compatible and tested with the [GLAMOR](#) system for advanced image effects combined with the volumetric clouds.

Sample [Video](#)



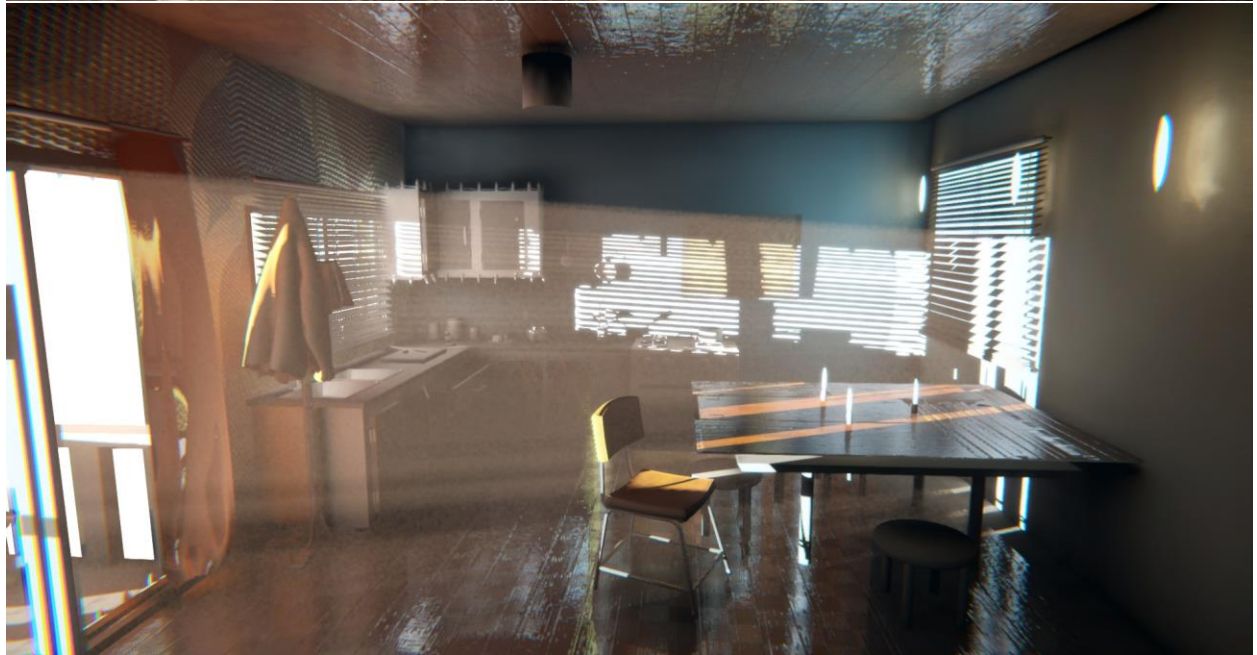
Use with LUMINA real time Global Illumination

The system has been extensively tested for compatibility with the [LUMINA real time Global Illumination](#) system, in both Unity 2022 and Unity 6 URP Render Graph mode. An example of use can be seen below, also demo for the functionality is available [here](#).

Demo (Windows – Use CTRL + Left Mouse Button to rotate sun): [Google Drive Link](#)

[Video 1](#) – [Video 2](#)



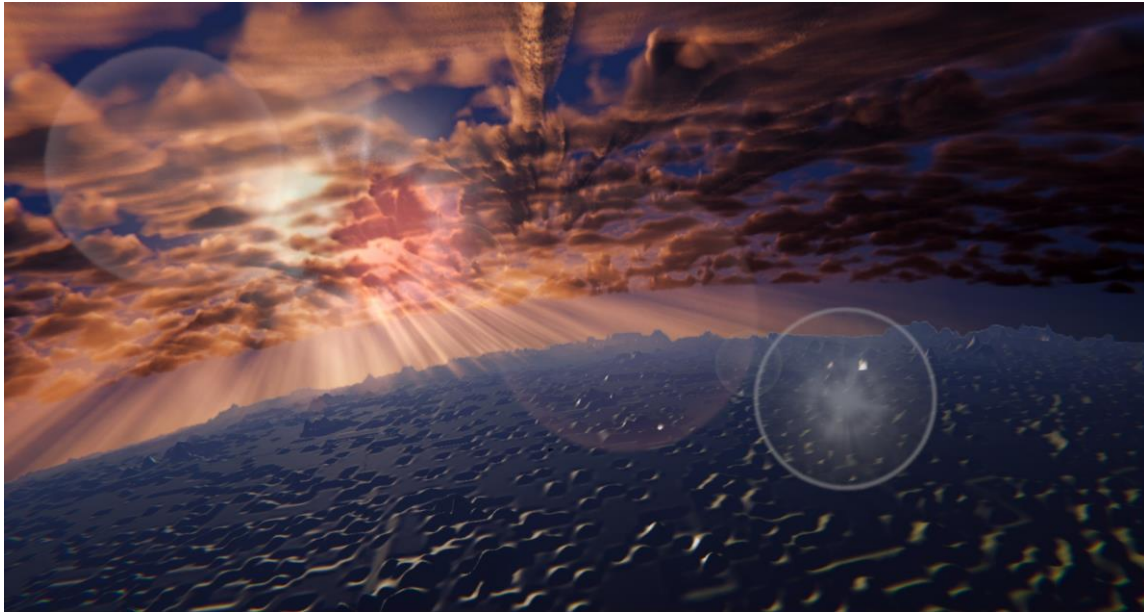


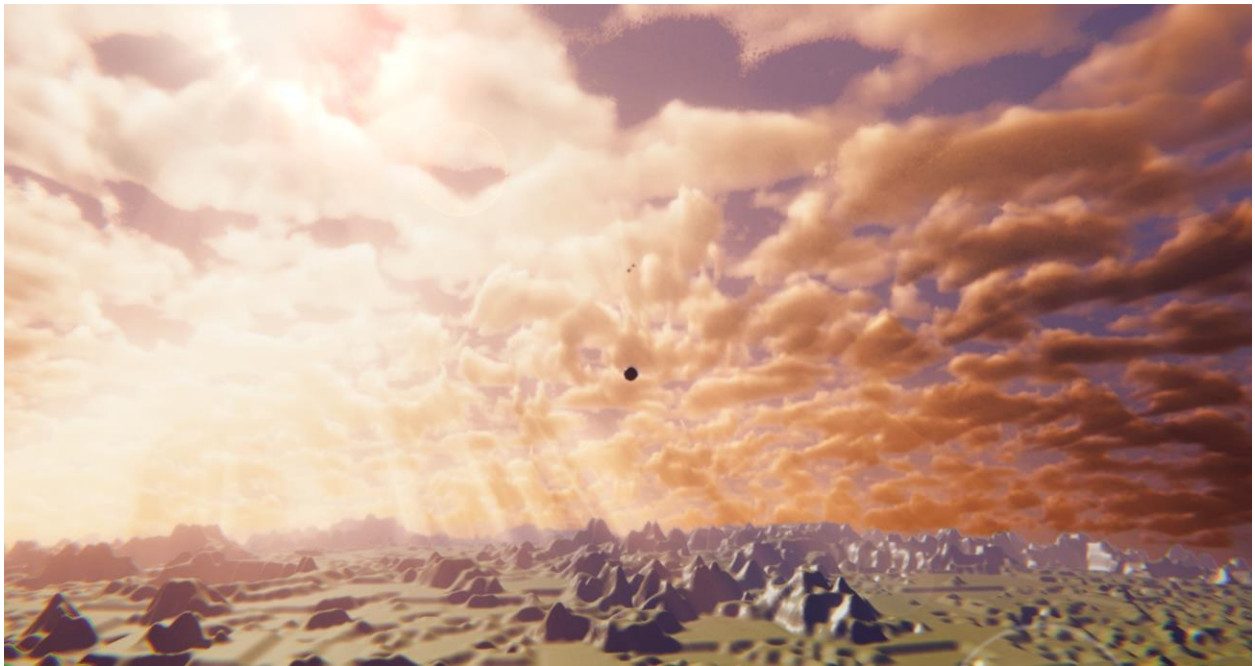
Other sun shafts options in Sky Master Volumetric Clouds & ORION space scene creation systems

The system can be combined with the various other volumetric sun shaft options available in other ARTnGAME assets, like [ORION Space scene creation framework](#) and **Sky Master ULTIMATE**. Some samples of the sun shaft options in various use cases like planets and [planetary](#) clouds are show below.

Ideally can choose the sun shafts from either Ethereal core or ORION – Sky Master systems and add use Ethereal 2024 new module for very fast shadowed local light volumetrics, for maximum efficiency.







The above images are from Sky Master ULTIMATE volumetric clouds and sun shafts, with ORION procedural planets and volumetric planetary atmosphere. The below image is from ORION Volumetric Nebula system, that can also can sun shafts in the Nebula volume.

